Probabilistic Calibration for MLP Uncertainty
Beyond Deterministic Spray Baselines

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The Core Argument

Goal: Show that the production MLP not only predicts accurate mean penetration trajectories but also provides **deliberately conservative dynamic uncertainty bounds (σ)**. The refreshed comparison keeps SVGP as the point-accuracy winner and frames the MLP as the more cautious screening surrogate.

The Limitation of Traditional Baselines:

- Physical baselines (Hiroyasu-Arai, Naber-Siebers) are fundamentally **deterministic**.
- They predict a point estimate, but any provided error boundaries are usually just static, globally computed historical residuals.
- They cannot dynamically adjust confidence based on the instantaneous out-of-distribution (OOD) nature of the specific injection condition.

The Solution: We upgrade our evaluation from simple empirical coverage metrics to formal **Probabilistic Forecasting Diagnostics**.

Evaluation Protocol

Current diagnostic sets

The refreshed full-clean CDF protocol has $692\,942$ points. The main fixed diagnostic table, `cdf_points_uncensored.csv`, has $542\,565$ points after conservative right-censoring removal. For every point i, diagnostics consume y_i, μ_i, and σ_i.

Why this matters

RMSE and MAE only test μ_i. Calibration diagnostics test whether σ_i is statistically meaningful: a useful uncertainty model must be both *calibrated* and *sharp*.

Models compared

Calibrated Hiroyasu–Arai, Naber–Siebers delay, production MLP anchor-off mean over five seeds, μ-anchor raw-uncertain MLP mean over five seeds, and the Stage 3 SVGP seed-42 baseline.

Source: `MLP/baseline/comparison_reports/stage3_fixed_table_eval_with_ha_ns_20260521/`

Metric 1: Reliability Diagram & Expected Calibration Error (ECE)

Step 1: convert each observation to a predictive percentile

$$u_i = F_i(y_i) = \Phi\left(\frac{y_i - \mu_i}{\sigma_i}\right),$$

where F_i is the predicted Gaussian CDF.

Step 2: compare nominal and empirical lower-tail probabilities

$$\hat{p}(\alpha) = \frac{1}{n}\sum_i \mathbf{1}\{u_i \le \alpha\}, \qquad \mathrm{ECE} = \frac{1}{|\mathcal{A}|}\sum_{\alpha\in\mathcal{A}} |\hat{p}(\alpha) - \alpha|.$$

- A calibrated model lies on the identity line $\hat{p}(\alpha) = \alpha$.
- ECE is unitless; lower is better.
- This is stricter than only checking $1\sigma/2\sigma$ coverage because it tests many probability levels.

Metric 2: Continuous Ranked Probability Score (CRPS)

Definition for a Gaussian predictive distribution

$$z_i = \frac{y_i - \mu_i}{\sigma_i}, \qquad \mathrm{CRPS}_i = \sigma_i \left[z_i(2\Phi(z_i) - 1) + 2\phi(z_i) - \frac{1}{\sqrt{\pi}} \right].$$

- CRPS has the same unit as the target: millimetres here.
- It is a proper scoring rule: the expected score is minimized by the true predictive distribution.
- Lower CRPS means a better calibration–sharpness trade-off, not merely a wider interval.
- Sharpness is reported separately as $\mathrm{mean}(\sigma_i)$; sharpness is good only when calibration remains acceptable.

Metric 3: Probability Integral Transform (PIT) & KS Test

PIT values are the same percentiles used for reliability:

$$u_i = \Phi\left(\frac{y_i - \mu_i}{\sigma_i}\right).$$

If the predictive distributions are calibrated, the set $\{u_i\}$ should be uniform on $[0, 1]$.

- U-shaped PIT: too many observations fall in predictive tails, often indicating under-dispersion / overconfidence.
- Hump-shaped PIT: too many observations fall near the predictive median, often indicating over-dispersion / conservative variance.
- A one-sided skew indicates systematic bias in μ.
- With $n = 692\,942$ on the full-clean table, KS p-values become extremely sensitive; histogram shape and ECE are more informative than treating $p = 0$ as a simple failure label.

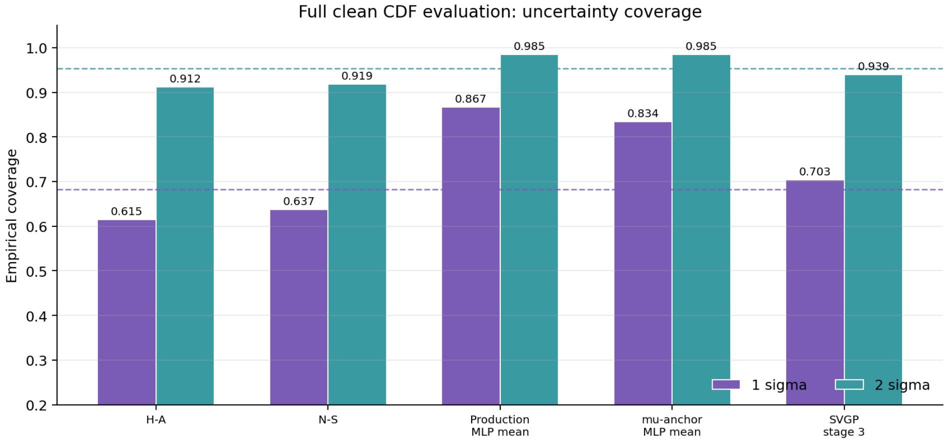
Current Coverage Summary

Model	Full RMSE	Full 1σ	Full 2σ	Uncens. RMSE	Uncens. 1σ	Uncens. 2σ
Hiroyasu–Arai calibrated	11.268	0.615	0.912	10.261	0.630	0.917
Naber–Siebers delay	10.545	0.637	0.919	9.286	0.659	0.927
Production MLP anchor-off mean(5)	4.735	0.867	0.985	4.265	0.887	0.989
μ-anchor raw-uncertain mean(5)	5.200	0.834	0.985	4.941	0.840	0.988
SVGP Stage 3	4.628	0.703	0.939	4.193	0.716	0.945

Readout: SVGP is the point-accuracy winner on both current tables. The production MLP is intentionally conservative: its $1\sigma/2\sigma$ coverage is well above the nominal 0.683/0.954 on the clean and uncensored sets.

Source: `Thesis/generated/baseline_comparison_20260521/fixed_table_group_summary.csv`

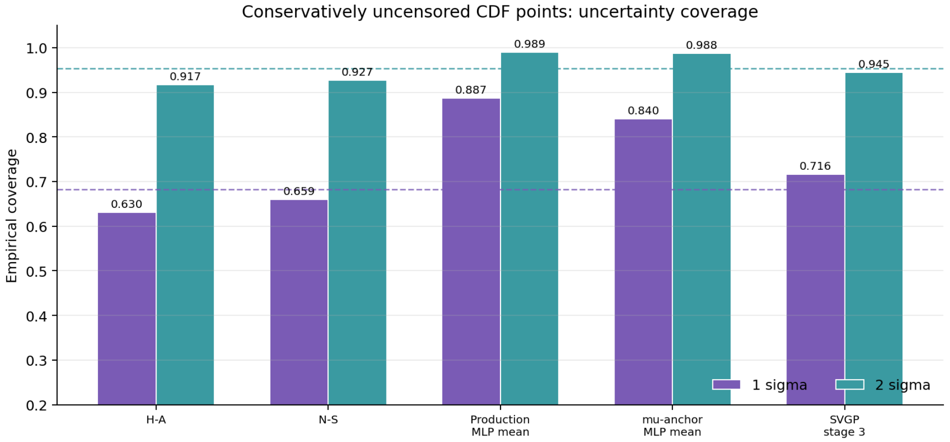
Full-Clean Coverage



Nominal Gaussian coverage is marked by dashed lines. Production MLP coverage is above nominal, while SVGP is closer to nominal but slightly under the 2σ target on the full-clean table.

Source: `Thesis/images/baseline_comparison_20260521/full_clean_coverage_comparison.png`

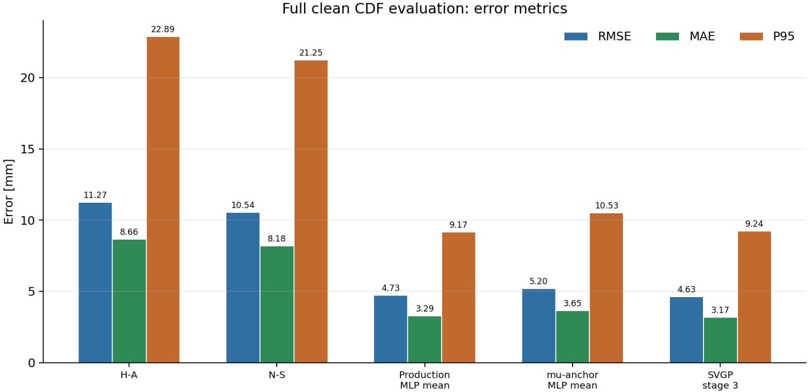
Uncensored-CDF Coverage



On the conservatively uncensored CDF table, production MLP remains over-covering and SVGP remains closer to nominal. This is the central accuracy–coverage trade-off used in the updated thesis narrative.

Source: `Thesis/images/baseline_comparison_20260521/cdf_uncensored_coverage_comparison.png`

Accuracy and Coverage Must Be Read Together



The updated message is balanced: MLP no longer claims to beat SVGP in point accuracy. It is close to SVGP, far ahead of HA/NS, and substantially more conservative in uncertainty coverage.

Source: `Thesis/images/baseline_comparison_20260521/full_clean_metric_bars_rmse_mae_p95.png`

Conclusion

Thesis Contribution: By integrating Reliability Diagrams, CRPS, and PIT histograms, we provide a complete suite of diagnostics (as implemented in `calibration_diagnostics.py`).

Takeaway:

Unlike Naber–Siebers or Hiroyasu–Arai models which output rigid curves plus global residual bands, the MLP acts as a heteroscedastic probabilistic surrogate. The current evidence is strongest in conservative coverage and deployment simplicity: the production five-seed MLP does not beat SVGP on RMSE, but it gives larger safety margins while staying far ahead of deterministic HA/NS baselines.